BT405 Nano biotechnology Assignment No.3 Solution

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Note: Rearrang relevant answer in ur own wording

Measures to control and overcome antibiotic resistance are urgently needed to avoid a clinical collapse which might be over the edge. Limiting the use of antibiotics while improving hygiene and antibiotic disposal measures have been the main action plans proposed by the World Health Organization (WHO) and governmental health agencies [4,5]. In parallel, significant funding is being allocated to the development of new and effective strategies against multidrug resistant (MDR) bacteria, with biological, adjuvant therapies, phages and small molecules with anti-quorum, anti-bacterial virulence or anti-toxin effects expanding the panel of potential therapeutic strategies [6]. However, the research and development of new antibiotics, biologicals or adjuvant therapies is a laborious process which cannot keep pace with the increasing rates of drug resistance and the urging need of fast-track approved approaches [7].

Taking this into account, an appealing alternative to the search of new therapies is the combination of currently approved antibiotics with the use of nanotechnology, also known as “nanobiotics”. Recent advances in this field have enabled the development of drug delivery systems with improved antimicrobial features and pharmacokinetic profiles [8,9,10]. These biomedical nanotechnology systems are remarkably improving the therapeutic effects of conventional drugs and may hold the promise of changing the efficacy of currently available antibiotics [11]. Among the wide array of nanoplatforms, one of the most promising delivery approaches for antibiotics under investigation are liposomes [12,13]. These lipid-based nano-systems were introduced as drug carriers in the 1970s, and, since then, major breakthroughs in liposome technology have driven the interest of their use as efficient delivery systems for antibacterial drugs [13,14,15].

The focus of this review aims to highlight the advantages of liposomes as carriers of antimicrobial agents and their potential to not only eradicate the infection but also overcome antibiotic resistance [16]. Hence, in an era of a massive increase of infections due to multidrug resistant bacteria, the use of antibiotic incorporated liposomes is a potential alternative to circumvent the limitations of conventional antimicrobial therapies [17,18].

Multiple nano-sized vesicles, such as metallic nanoparticles, liposomes, dendrimers, polymeric nanoparticles and carbon nanotubes, have been designed with improved therapeutic properties of the incorporated compounds, such as controlled release, decreased systemic toxicity, drug-targeting and higher efficiency [19].

Within these nanotechnological-based approaches available, lipid-based nano-systems such as liposomes have demonstrated particularly appealing features in terms of physicochemical properties and safety issues. Liposomes are vesicular concentric bilayer structures composed of relatively biocompatible and biodegradable materials. They offer several advantages over other delivery systems due to their unique characteristics to incorporate hydrophilic and hydrophobic drugs, biocompatibility, biodegradability, low toxicity and lack of immune system activation [14,20,21]. In addition, liposomes can be easily coupled with targeting platforms, such as antibodies, proteins or enzymes, thus allowing a specific delivery of bioactive compounds directly into infection sites [22,23,24,25]. More importantly, as detailed in Table 1, several liposomal-based formulations have been approved by the FDA for clinical use or are in clinical trials in the infectious disease setting, demonstrating their therapeutic potential and the possibility of fast-track approval of subsequent formulations.

Liposomal formulations are generally characterized by size (small unilamellar vesicles (SUVs) < 100 nm, large unilamellar vesicles (LUV) > 100 nm), number of lamellae (unilamellar or multilamellar vesicles), lipid composition, charge of the bilayer (anionic, cationic or neutral) and surface functionalization with polymers or ligands. These properties are known to influence their stability and biological performance both in vitro and in vivo [25,29]. Charge and composition are particularly important features since they arbitrate the fluidity and stability of the liposomal membrane and impact the liposome–bacteria interaction [20,21,26,29]. An essential characteristic that upraises liposomes to potential disruptive antibacterial therapeutics is, as more deeply discussed below, their flexibility for surface modification. The surface functionalization, as detailed in Figure 1, with ligands, such as polymers (e.g., PEGylated liposomes) and molecules (e.g., antibodies, proteins/peptides and carbohydrates), is used for specific targeting (ligand-targeted liposomes) [12,29], playing an important role in efficient delivery and therapeutic efficacy.

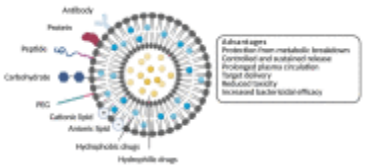


Figure 1

Schematic representation of different types of liposomes and their major advantages.

4. Advantages of Liposomes as Antibiotic Carriers

Recent improvements in liposomal formulations have enabled the development of potential antibiotic delivery platforms that could revise critical issues in the treatment of infectious diseases. As described above, liposomes have several advantages as antibiotic delivery nanosystems, overcoming the problems related either with efficacy of the incorporated drug or selection of resistant strains. Several studies have shown that liposomal encapsulation promotes the stability and safety of antibiotics, giving rise to more appropriate pharmacokinetic and pharmacodynamic profiles by prolonging the circulation time in bloodstream, enabling a specific targeting to the infection sites using different routes of administration (Figure 2) [22,30].



Figure 2

Schematic representation of the main advantages of liposomes as antibiotic carriers.

4.1. Stability

4.1.1. Controlled and Sustained Release of Antibiotics

Bioavailability and antibiotic concentration at the infection site is highly dependent on the administration route (intravenous, oral and pulmonary), class of antibiotics, dosing and treatment duration, drug interaction, co-morbidities and overall patient condition (particularly in critically ill patients) [31,32,33]. This variability impacts

not only the infection clearance rate but also may contribute to the development of antimicrobial resistance since only a fraction of the antibiotic actually reaches its target. For this reason, when free drugs are administered, high and repeated dosing is indispensable to maintain antibiotic concentrations above the bacteria minimum inhibitory concentration (MIC) [33].

Nevertheless, several studies have demonstrated that, for drugs that present concentration dependent activity, such as fluoroquinolones and aminoglycosides, higher antibiotic dosages are more efficient in clearing and preventing the selection of resistant strains than fractionating regimens [34,35]. However, the use of high concentrations is limited by dose-dependent toxic effects of the antibiotics, which makes the management of the therapeutic dosing of an infection, caused by an antimicrobial resistant, a clinical challenge [32].

A major advantage of incorporating antibiotics in liposomes is the possibility to regulate the release of the entrapped antibiotic. Depending on their composition and presence of specific stimulatory factors, such as pH or heat, liposomes can be engineered to disintegrate and subsequently release incorporated drugs in a controlled manner [21,29]. This may occur only at the infection site, without premature release during circulation or be sustained over a period of time enabling the reduction of dosing frequency and consequent systemic toxicity [13,36,37,38]. The sustained release of antibiotics may be, by itself, highly beneficial against antimicrobial resistant strains since it enables the maintenance of a higher than MIC concentration without secondary toxicity